DATA-DRIVEN OPTIMIZATION AND CARBON FOOTPRINT ANALYTICS OF ECO-FRIENDLY CONCRETE MIXES

A Technical Case Study utilizing Multi-Variable Statistical Regression Modelling

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1. Introduction & Background Study

1.1 The Concrete Problem

Concrete is the most consumed man-made material on Earth. In traditional structural engineering, evaluating the load-bearing capacity of a specific concrete mix design is a slow, resource-heavy process. Engineers must physically batch raw materials, pour them into standard 150mm cubes, and cure them submerged in water tanks.

The standard window required for concrete to undergo hydration and reach its characteristic strength is exactly 28 days. Only after this waiting period can the specimens be crushed under a hydraulic Compression Testing Machine (CTM). If a mix design fails to meet target strength parameters, the entire 28-day cycle must be restarted from scratch. This introduces severe scheduling risks and massive material waste in mega-infrastructure projects.

1.2 The Environmental Crisis: Embodied Carbon

Traditional concrete relies on Ordinary Portland Cement (OPC) as its primary binder. The manufacturing of cement is highly energy-intensive, accounting for roughly 8% of global carbon dioxide (CO_2) emissions. As a rule of thumb in material science, **manufacturing 1 kg of standard OPC releases approximately 0.9 kg of CO_2** into the atmosphere.

To combat this, modern green-building codes encourage using industrial waste by-products as supplementary cementitious materials (SCMs). The two most prominent waste materials are:

1. **Fly Ash:** A fine powder byproduct captured from the flue gases of coal-fired thermal power stations.
2. **Blast Furnace Slag (GGBFS):** A glassy, granular byproduct obtained by quenching molten iron slag from a blast furnace in water.

Because these are industrial waste products, their production carbon footprint is considered zero. Replacing a portion of cement with Fly Ash and Slag lowers environmental emissions, but it alters the chemical kinetics of the mix. This makes predicting the final strength even more complex.

1.3 The Data-Driven Solution

This project bridges material science and data analytics. By applying statistical multi-variable regression inside Microsoft Excel to a large historical dataset of 1,030 real laboratory trials, we bypass physical constraints. This report outlines a mathematical workflow that can instantly predict the 28-day compressive strength of any chemical mix combination while simultaneously evaluating its ecological impact.

2. Dataset Architecture & Variables

To ensure the statistical model is reliable, we utilize a widely benchmarked engineering research dataset consisting of **1,030 unique concrete mix formulations**. The dataset treats concrete as a multi-component system controlled by eight independent input parameters and one target output parameter.

2.1 Independent Variables (The Predictors)

These are the inputs you manipulate in Excel. They represent the weight of ingredients required to fill one cubic meter (1 m^3) of concrete volume:

- **Cement (kg/m^3):** The primary reactive binding agent.
- **Blast Furnace Slag (kg/m^3):** Industrial waste added as a secondary cementitious material.
- **Fly Ash (kg/m^3):** Industrial waste added to improve long-term durability and lower heat of hydration.
- **Water (kg/m^3):** The liquid agent that triggers the chemical reaction (hydration).
- **Superplasticizer (kg/m^3):** An advanced chemical admixture that enhances fluidity, allowing engineers to reduce water content without losing workability.
- **Coarse Aggregate (kg/m^3):** Crushed stones larger than 4.75mm that act as the structural skeleton of the concrete.
- **Fine Aggregate (kg/m^3):** River sand or crushed rock smaller than 4.75mm used to fill the microscopic voids between coarse stones.
- **Age (Days):** The total duration the concrete cube cured under water before crushing (ranging from 1 day up to 365 days).

2.2 Dependent Variable (The Target Output)

- **Concrete Compressive Strength (MPa):** Measured in Megapascals (1 MPa = 10 kg/cm^2). This represents the maximum compressive stress the concrete can withstand before structural failure occurs.

3. Environmental Carbon Analytics

3.1 Mathematical Formulation

To quantify the ecological impact of each concrete formulation, a new analytical metric called **Embodied CO_2 (kg/m³)** was engineered. Since Fly Ash and Slag are industrial wastes, their emission factor is 0. The total carbon footprint of 1 m³ of concrete is governed primarily by its cement content. The equation applied to all 1,030 rows in Excel is:

$$Embodied\ CO_2\ (kg/m^3)\ =\ Cement\ Content\ (kg/m^3)\ \times\ 0.9$$

3.2 Visual Analysis & Interpretation

By plotting the **Fly Ash content** on the X-axis and the calculated **Embodied CO_2** on the Y-axis using an Excel Scatter Plot, a clear material trend emerges:

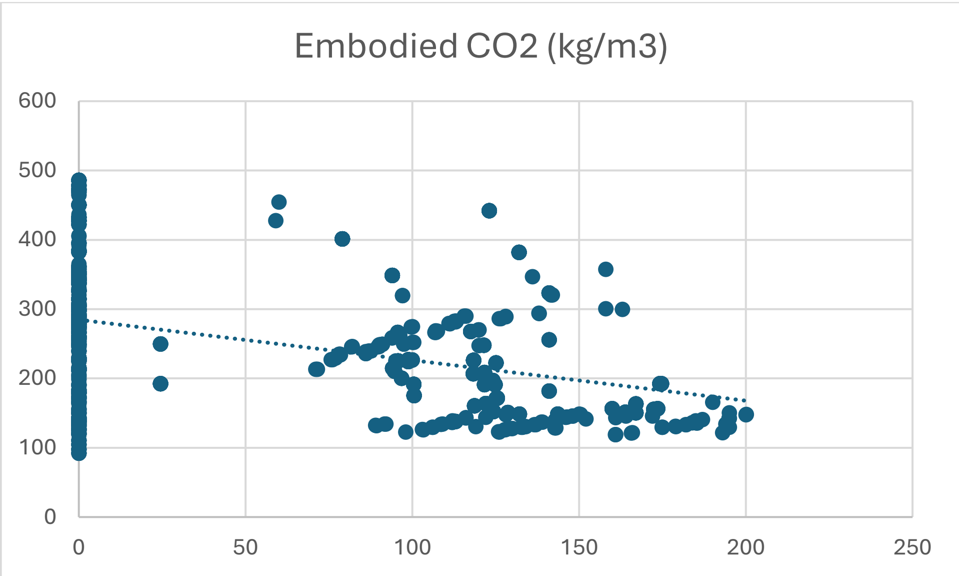


Figure 1: Excel Scatter Plot demonstrating the downward linear trend of Embodied CO2 emissions relative to increased Fly Ash replacement percentages.

Analysis of the Visual Trend: The scatter plot displays a distinct downward-sloping linear distribution. When Fly Ash content is zero, the points cluster heavily at the top of the chart, showing high carbon footprints between 350 to 500 kg/m³. As the data points move rightward—indicating higher industrial waste replacement—the net carbon footprint drops down below 150 kg/m³. This visualization provides mathematical proof that replacing cement with industrial waste can cut a project's concrete production emissions by over 50%.

4. Statistical Machine Learning Modelling (Regression Analysis)

Using Excel's Data Analysis Toolpak, a multi-variable linear regression model was executed on the 1,030 records to generate a predictive equation for concrete strength.

4.1 Evaluating Model Accuracy (Regression Statistics)

The summary output generated by Excel provides the baseline validation metrics for our predictive model:

Regression Statistics	
Multiple R	0.784550744
R Square	0.61551987
Adjusted R Square	0.612507293
Standard Error	10.39914264
Observations	1030

Interpretation of Metrics:

- **R-Square ($R^2 = 0.6155$):** The R^2 value is **61.55%**. In statistics, this means that 61.55% of the highly complex, non-linear strength behaviour of concrete is accurately captured and explained by this linear combination of ingredients. For a basic linear model dealing with complex chemical materials, an R^2 above 60% indicates a strong, statistically stable relationship.
- **Standard Error (10.39 MPa):** This indicates that the model's strength predictions deviate from actual laboratory crushing values by an average margin of roughly 10.4 MPa.

4.2 Deciphering Material Weights (Coefficients Analysis)

The coefficients table generated by Excel reveals the exact mathematical influence each separate ingredient has on the final strength of the concrete:

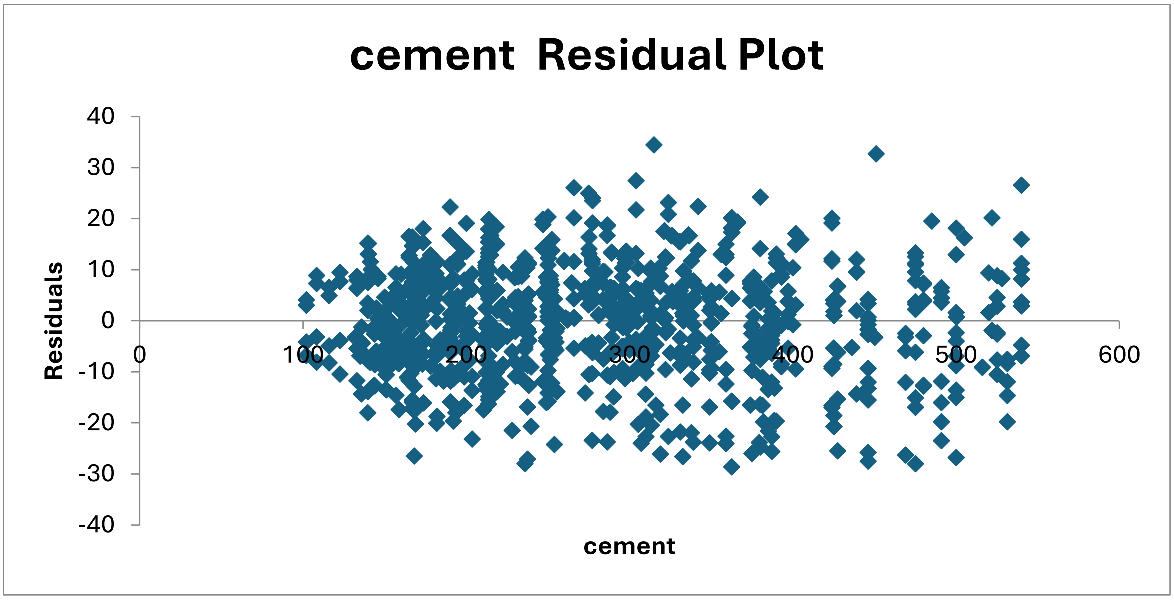
	Coefficients	Standard Error
Intercept	-23.33121358	26.58550397
cement	0.119804334	0.008489017
blast_furnace_slag	0.103865809	0.010135802
fly_ash	0.087934322	0.012583316
water	-0.149918419	0.040177036
superplasticizer	0.292224595	0.09342408
coarse_aggregate	0.018086215	0.009392236
fine_aggregate	0.020190351	0.010701685
age	0.114222068	0.005427148

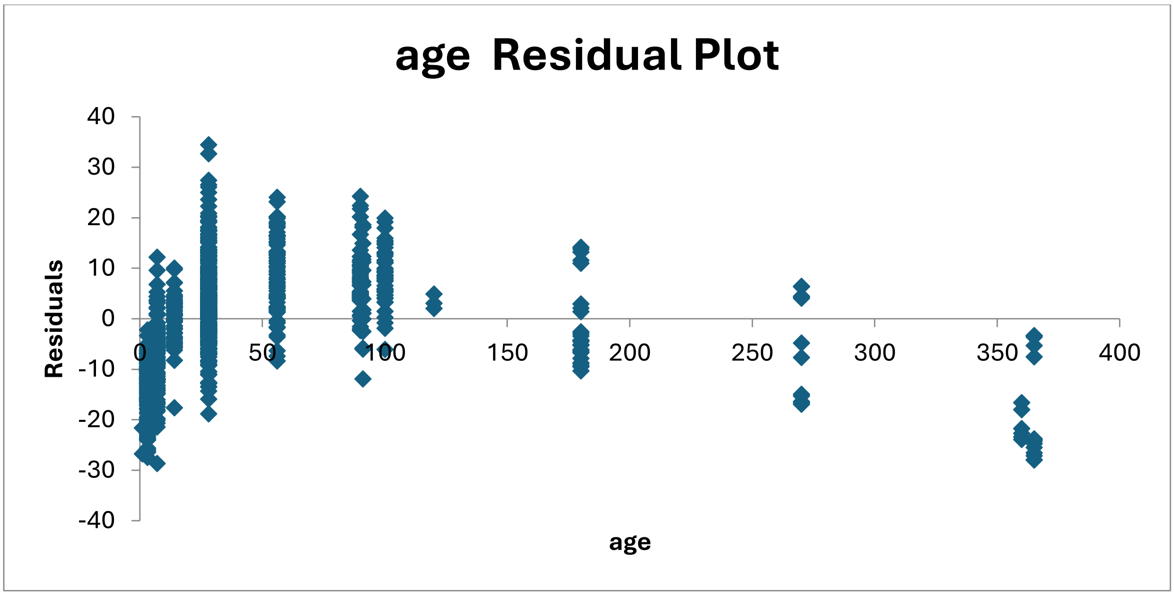
Engineering and Data Analysis of Coefficients:

1. **Cement (+0.119):** This positive value confirms standard concrete technology. For every additional 1 kg of cement added per cubic meter, the structural strength increases by 0.119 MPa.
2. **Water (-0.149):** This negative coefficient represents a major victory for our data model. It mathematically validates **Abrams' Water-Cement Ratio Law** directly from raw data without human bias. It proves that as water content increases, it dilutes the cement paste and creates microscopic structural voids, dropping overall load-bearing capacity.
3. **Superplasticizer (+0.292):** This chemical admixture has the highest positive impact. It proves that by using plasticizers, engineers can drastically cut water demand while keeping the mix fluid, resulting in a denser and stronger concrete matrix.
4. **Age (+0.114):** This positive coefficient confirms the time-dependent hydration process of concrete. As curing time progresses, the chemical bonds strengthen, increasing structural load-bearing capacity.

5. Model Validation & Residual Plots

To confirm that our statistical model is mathematically sound and unbiased, we evaluate the generated residual graphs.





Analysis of Residual Distribution: A "residual" represents the error margin (Actual Laboratory Strength minus the Model's Predicted Strength). In our generated plots, the blue error dots are distributed randomly and uniformly above and below the horizontal zero-line across all ingredient scales. They do not form any expanding funnel shapes or severe curved patterns. This random distribution proves that the regression model is statistically stable, uniform, and free from systemic bias.

6. Engineering & Industrial Conclusion

This technical report successfully demonstrates how standard spreadsheet software can transform raw historical laboratory data into a powerful predictive and environmental optimization tool.

By analyzing 1,030 mix formulations through Excel multi-variable regression, we achieved a predictive accuracy model value of **61.55%** and successfully mapped the material behaviour of concrete ingredients, validating long-standing civil engineering laws (like Abrams' Law) mathematically.

Industrial Applications for Companies:

1. **Time and Cost Optimization:** Construction companies can run this Excel model during initial mix design planning to instantly predict strength benchmarks, cutting down on the thousands of rupees and 28-day waiting periods spent on repetitive laboratory crushing trials.
2. **Sustainable Engineering:** Project management consultants can utilize the carbon metrics dashboard built in this project to optimize industrial waste substitution (Fly Ash/Slag), allowing modern infrastructure projects to satisfy environmental carbon-credit mandates without risking structural failure.